Dynamic Material Balance—Oil- or Gas-in-Place Without Shut-Ins

L. MATTAR, D. ANDERSON, G. STOTTS
Fekete Associates Inc.

Abstract

Material balance calculations for determining oil- or gas-in-place require static reservoir pressures, which can only be obtained when the well is shut in. In a previous publication[1] titled "The 'Flowing' Gas Material Balance," it was shown that the reservoir pressure could be obtained from the flowing pressure for wells producing at a constant rate.

The "Dynamic Material Balance" is an extension of the "Flowing Material Balance" and can be applied to either constant or variable flow rates. Both methods are applicable for gas and oil. The "Dynamic Material Balance" is a procedure that converts the flowing pressure at any point in time to the average reservoir pressure that exists in the reservoir at that time. Once that is done, the classical material balance calculations become applicable, and a conventional material balance plot can be generated.

The procedure is graphical and very straightforward: a) knowing the flow rate and flowing sandface pressure at any given point in time, convert the measured flowing pressure to the average pressure that exists in the reservoir at that time; and, b) use this calculated average reservoir pressure and the corresponding cumulative production, to calculate the original oil- or gas-in-place by traditional methods. The method is illustrated using data sets.

Introduction

The material balance method is a fundamental calculation in reservoir engineering, and is considered to yield one of the more reliable estimates of hydrocarbons in place. In principle, it consists of producing a certain amount of fluids, measuring the average reservoir pressure before and after the production, and with knowledge of the PVT properties of the system, calculating a mass balance as follows:

Remaining hydrocarbons-in-place = initial hydrocarbons-in-place – produced hydrocarbons

At face value, the above equation is simple; however in practice, its implementation can be quite complex, as one must account for such variables as external fluid influx (water drive), compressibility of all the fluids and of the rock, hydrocarbon phase changes, etc.

In order to determine the average reservoir pressure, the well is shut in, resulting in loss of production. In high permeability reservoirs, this may not be a significant issue, but in medium to low permeability reservoirs, the shut-in duration may have to last several weeks (and sometimes months) before a reliable reservoir pressure can be estimated. This loss of production opportunity, as well as the cost of monitoring the shut-in pressure, is often unacceptable.

It is clear that the production rate of a well is a function of many factors such as permeability, viscosity, thickness, etc. Also, the rate is directly related to the driving force in the reservoir, i.e., the difference between the average reservoir pressure and the

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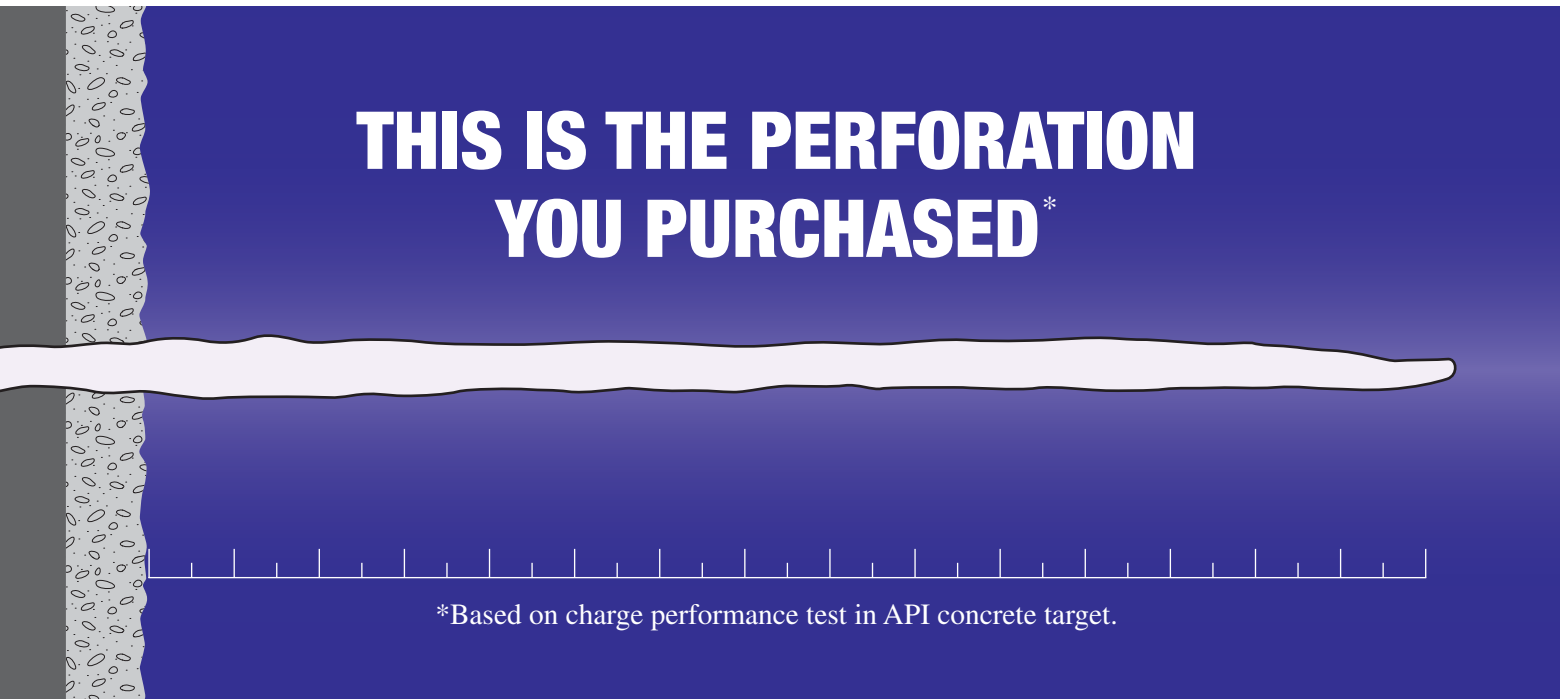

sandface flowing pressure. Therefore, it is reasonable to expect that knowledge about the reservoir pressure can be extracted from the sandface flowing pressure if both the flow rate and flowing pressure are measured. If, indeed, the average reservoir pressure can be obtained from flowing conditions, then material balance calculations can be performed without having to shut in the well. This is of great practical value. In a previous publication[1], the authors obtained a relationship between the well flowing pressure (which can be measured) and the average reservoir pressure. They called that procedure "The Flowing Material Balance." In this Technical Note, that procedure is extended to situations where the flow rate is not constant. It is called the variable rate flowing material balance or "Dynamic Material Balance."

Details of the flowing material balance and the dynamic material balance can be found in References 1 and 2. The equations are derived for a "volumetric" reservoir (i.e., no water drive or external fluid influx), but the method can be extended to include such complexities. The method is valid for both oil and gas systems, but it is sometimes more convenient to present a particular concept (or equation) in terms of gas rather than oil, or vice versa.

Dynamic Material Balance (Variable Rate Flowing P/Z Plot)

The flowing material balance is restricted to constant rate production. As a well produces at a constant rate in pseudo-steady state flow, there is a consistent difference between the sandface flowing pressure and the average reservoir pressure. This relationship can be used to acquire the average reservoir pressure and construct the material balance plot[1]. However, many wells incur significant variations in rate and flowing pressure over their production life. The dynamic material balance[2] is applicable to both constant rate and variable rate production. The complete development of the appropriate equations can be found in References 2 and 3. A simplified summary of the concepts as they apply to variable rate production is presented below:

Pseudo-steady state flow:

$$p_i - p_{wf} = \frac{qt}{c_o N} + b_{pss} q \qquad (1)$$

Cumulative production:

$$\left(q \times t = N_p\right) \qquad (2)$$

Material balance equation:

$$p_i - \bar{p}_R = \frac{N_p}{c_o N} \qquad (3)$$

Combining Equations (1), (2), and (3):

$$\bar{p}_R - p_{wf} = b_{pss} q \qquad (4)$$

Re-arranging:

$$\bar{p}_R = p_{wf} + b_{pss} q \qquad (5)$$

The above equation illustrates how the dynamic material balance can be applied to a well with a varying production rate and a corresponding varying flowing pressure. The conversion from flowing pressure to average reservoir pressure must take into account the varying flow rate. Since the flow rate is known, we need only determine the value of b_{pss}, using some independent method. A plot of $(p_i - p_{wf}/q)$ vs. N_p/q should yield a straight line when boundary dominated flow is reached, as shown in Figure 1. The intercept of this plot is b_{pss}. Note that the value of b_{pss} is subject to interpretation, as it depends on the proper identification of the stabilized (straight-line) section of the graph.

The above summary equations apply to a single phase liquid system. Appendix C of CIPC 2005-113[2] presents the corresponding equations for a gas reservoir. For a gas reservoir, two modifications are necessary:

a) The pressure must be converted to pseudo-pressure to account for the dependence of viscosity and Z-factor on pressure[4]; and,
b) Material balance time[5-7] must be converted to pseudo-time to account for the strong dependence of gas compressibility on pressure.

The step-by-step procedure for generating a dynamic material balance plot for a gas well with varying flow rate is given below:

1. Convert initial pressure to pseudo-pressure, p_{pi};
2. Convert all flowing pressures to pseudo-pressures, p_{pwf};
3. Assume a value for the original gas-in-place, G;
4. Calculate pseudo-time from Equation (C-11)[2], t_{ca};
5. Plot $(p_{pi}-p_{pwf}/q)$ vs. pseudo-time, t_{ca}. The intercept gives b_{pss}. See Figure 1;
6. Calculate the average reservoir pseudo-pressure from Equation (C-19)[2];
7. Convert the average reservoir pseudo-pressure to average reservoir pressure, p_R;
8. Calculate p_R/Z and plot against cumulative gas produced, Gp, just like the conventional material balance graph for a gas pool. The intercept on the X-axis gives the original gas-in-place, G. See Figure 2; and,
9. Using this new value of G, repeat Steps 3 to 7 until G converges.

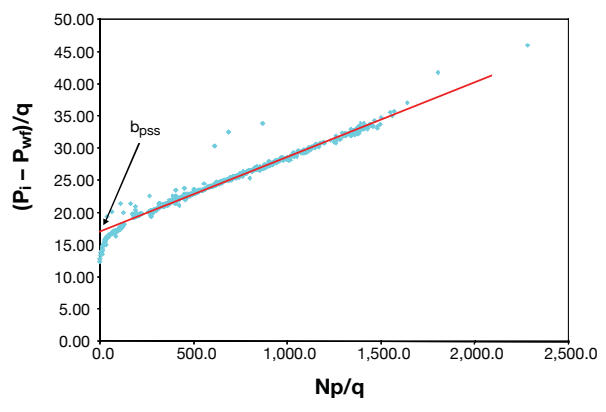

FIGURE 1: Determination of b_{pss}.

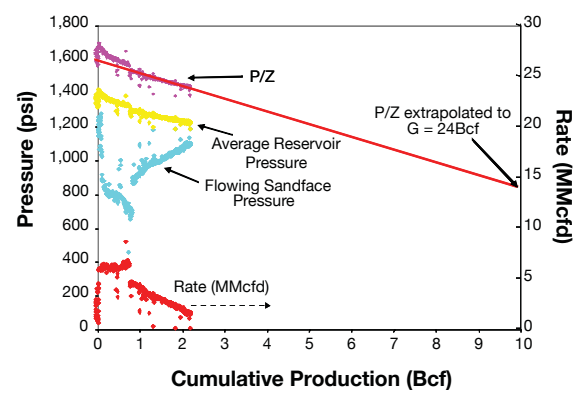

FIGURE 2: Dynamic material balance plot.

Limitations

The procedure described in this paper is very effective and provides extremely valuable information. However, like any other reservoir engineering, it has its limitations:

- Since material balance time and pseudo-time are rigorous only during boundary-dominated flow, data obtained during transient flow cannot be used in this analysis. The transient data can be identified as the curved part of the graph in Figures 1 and 2, and should be ignored;
- In certain situations such as pressure-dependent permeability, or continuously changing skin (both factors have been ignored in the development of the equations), this method will tend to under-predict the hydrocarbons-in-place. These factors can be accounted for by more complex definitions of pseudo-pressure and pseudo-time; and,
- The dynamic material balance is an "indirect" method of determining the average reservoir pressure. As such, it incorporates many assumptions. On the other hand, build-up tests themselves have their own sets of assumptions when the build-up pressure has to be extrapolated to obtain the average reservoir pressure. Accordingly, whenever possible, these methods should be used in concert with each other rather than as alternatives to each other.

Conclusion

- It is possible to obtain the average reservoir pressure without shutting in a well.
- The flowing pressure can be converted to the average reservoir pressure existing at the time of the measurement using a very direct procedure.
- The average reservoir pressure obtained from the dynamic material balance method can be used anywhere the average reservoir pressure has traditionally been used.
- For a gas well, a conventional p_R/Z plot can easily be generated without shutting in the well, and the original gas-in-place determined as usual.
- The dynamic material balance applies to variable rate production.
- The dynamic material balance should not be viewed as a replacement to build-up tests, but as a very inexpensive supplement to them.

NOMENCLATURE

c_o = oil compressibility

b_{pss} = $\dfrac{141.2B\mu}{kh}\left[\ln\left(\dfrac{r_e}{r_{wa}}\right)-\dfrac{3}{4}\right]$ (field units)

b_{pss} = $\dfrac{11.57B\mu}{kh}\left[\ln\left(\dfrac{r_e}{r_{wa}}\right)-\dfrac{3}{4}\right]$ (metric units)

G = original gas-in-place
G_p = cumulative gas produced
h = pay thickness
k = reservoir permeability
N = original oil-in-place
N_p = cumulative production produced
p_i = initial reservoir pressure
$\bar{P}_R$ = average reservoir pressure
p_{wf} = flowing pressure
p_p = pseudo-pressure
p_{pi} = pseudo-pressure at initial reservoir pressure
p_{pwf} = pseudo-pressure at the flowing pressure
q = production rate (can be a function of time)
r_e = exterior radius
r_{wa} = apparent wellbore radius
t = time

t_{ca} = material balance pseudo-time for gas = $\displaystyle\int \frac{dt}{\mu\,\bar{c}_g}$

T = reservoir temperature, R°
$\bar{Z}$ = compressibility factor at average reservoir pressure
B = oil formation volume factor
μ = viscosity

SI Conversion Factors

1 psia = 6.895 kPa
1 MMscfd = 28.32 $10^3 m^3/d$
1 bcf = 28.32 $10^6 m^3$

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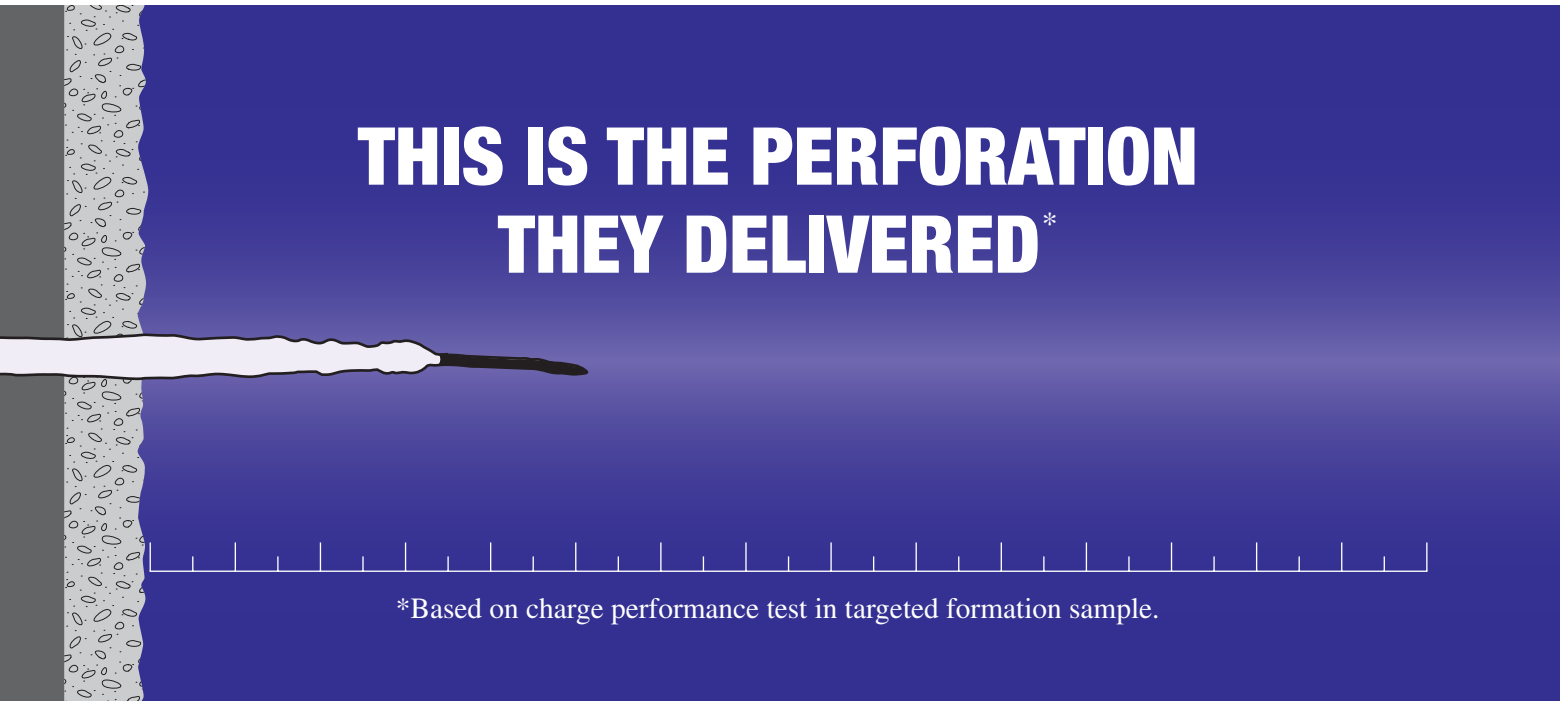

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Authors' Biographies

Louis Mattar is the president of Fekete Associates Inc. He was the principal author of the world-renowned E.R.C.B. publication, "Theory & Practice of the Testing of Gas Wells, 1975." He specializes in well testing and teaches it all around the world. He has authored 45 technical publications. He is a distinguished member of the Petroleum Society. In 1995, he received the Petroleum Society Distinguished Author Award, and the Outstanding Service Award. In 2003, Louis was the SPE distinguished lecturer in well testing.

David Anderson (P.Eng) is a technical advisor with Fekete Associates Inc. He has eight years of experience in the petroleum industry, including production optimization, gas deliverability modelling, and well test analysis. He is currently the technical leader for Fekete's RTA (Rate Transient Analysis) Group. He has taught numerous industry courses on advanced production decline analysis and has co-authored several technical publications on both pressure and rate transient analysis.

Garth Stotts (E.I.T.) is a project engineer at Fekete Associates Inc. He acquired a B.Sc. (with Distinction) in materials engineering from the University of Alberta, and is currently working toward a M.Eng. at the University of Calgary.

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